

Shannon Dooley

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EDUCATION

University of Pennsylvania, School of Engineering and Applied Science

Philadelphia, PA | May 2027

Candidate for BSE and MSE in *Systems Science and Engineering*

Concentration: *Data Science and AI*

Minors: *Engineering Entrepreneurship, Mathematics*

PROFESSIONAL EXPERIENCE

Wharton Venture Initiation Program

Founder

September 2025 – Present

- Led early-stage venture development from idea validation through MVP scoping, with a focus on customer discovery and founder-market fit
- Evaluated market opportunity, competitive dynamics, and operational risks to inform go-to-market and capital planning decisions.

Barclays Investment Bank

New York, NY | June – August 2025

Cyber and Technology Audit Summer Analyst

- Evaluated firm-wide technology and control risks across large-scale financial systems, supporting senior stakeholders in risk-based decision-making.
- Built and deployed a Python-based analytics pipeline to assess endpoint control effectiveness, improving identification of coverage gaps and operational failure modes.
- Synthesized complex technical findings into concise, decision-ready materials for internal reviews and regulatory discussions.

DARPA Triage Challenge

Philadelphia, PA | February – December 2024

AI Research Intern

- Owned end-to-end analytical workflows under strict time and performance constraints, balancing accuracy, speed, and operational risk.
 - Designed data ingestion and preprocessing pipelines to handle noisy, high-frequency time-series sensor data under strict latency constraints.
 - Worked cross-functionally with clinicians and engineers to translate abstract requirements into executable systems.
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LEADERSHIP & EXTRACURRICULARS

Class Board

Engineering Chair for Class of 2027

September 2023 – Present

- Led campus-wide initiatives for 400+ engineering students, managing a \$50k budget and making strategic trade-offs across competing priorities.

Teaching Assistant

ESE 2030 (Linear Algebra with Applications in Engineering and AI)

January 2025 – Present

- Led problem-solving sessions and 1:1 instruction on linear algebra, eigenvalues, and matrix applications in AI and systems modeling.
 - Supported professors in grading and designing quantitative assignments, ensuring technical rigor and clarity.
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SKILLS & INTERESTS

Technical: Excel, Python, SQL, Java

Quantitative: Probability, Optimization, Stochastic Modeling, Numerical Simulation

Languages: Chinese (intermediate)

Interests: Capital Markets, Structured Products, Portfolio Construction, Private Markets, Venture Capital